

EAR PRO NRR

The number on the package is a lab score — not a gunshot rating. This is what it means on a SoCal firing line.

85 dB

HEARING DAMAGE FROM STEADY NOISE STARTS AROUND HERE WITH ENOUGH TIME. A GUNSHOT IS A DIFFERENT ANIMAL — TYPICALLY 155-168 DB PEAK. UNPROTECTED PEAKS ABOVE 140 DB ARE ALREADY OVER THE OCCUPATIONAL CEILING.

01 GUNSHOT VS. RANGE VS. THAT 85 DB NUMBER

SHOP-FLOOR 85

NIOSH's 8-hour recommended limit is 85 dBA. OSHA's legal 8-hour cap is 90 dBA; they start a hearing program at 85. Those are averages over a shift — grinders, HVAC, traffic.

INDOORS ADDS BOUNCE

Walls throw the blast back. Your neighbor's muzzle and the concrete are part of your dose. That is why a range posts "ear pro required" and never bothers printing 85 — the number on a construction poster is not the number at the firing line.

A SHOT IS IMPULSE

Pistol and rifle peaks at the ear commonly land 155–168 dB. Some rifles and indoor bays run higher. OSHA and NIOSH both say: do not take unprotected impulse peaks above 140 dB.

02 FOAM VS. MUFFS VS. ELECTRONIC

FOAM

LABELED NRR ~29-33

Highest number on the package — *if* they go in. Roll into a crease, pull the ear up and back, insert, hold until they expand. A plug sitting in the bowl of the ear is jewelry, not protection. Muffles commands.

WHEN

You want max attenuation. Under muffs indoors. You will insert them correctly.

MUFFS

LABELED NRR ~20-31

Consistent. You can see they are on. No insertion skill required. Glasses, hats, and hair break the seal — stems under the cups leak. Easy on and off between strings.

WHEN

You want honest, visible coverage. Glasses with thin stems, or wear the frames over the cushion.

ELECTRONIC

LABELED NRR OFTEN ~20-26

Lets you hear "cease fire" and the person next to you, then clamps down on the shot. The electronics are awareness, not extra NRR — the cups are the protection. Dead batteries = dumb muffs. Not a toy.

WHEN

You need commands and still have enough cup NRR for the gun. Best with foam underneath indoors.

03 WHEN TO DOUBLE UP INDOORS

1 INDOOR RIFLE — YES.

Foam plus muffs. NIOSH tells indoor-range workers to dual-protect whenever the line is hot. Magnum, short bay, or a crowded indoor pistol line: same rule.

3 DO NOT ADD THE TWO NRRS.

Extra protection is about +5 dB over the better of the two — not $29+25=54$. Bone conduction is the ceiling.

2 NEED TO HEAR COMMANDS?

Foam under electronic muffs. That is the instructor setup — awareness on top, plug in the canal.

4 OUTDOOR, WELL-FITTED SINGLE DEVICE CAN COVER A QUIET PISTOL STRING.

Indoor, rifle, or a hot neighbor: double up. If in doubt, double up.

04 WHY THE PACKAGE NUMBER IS OPTIMISTIC

NRR is an EPA lab score on a well-trained test subject. Real ears, real glasses, real haste at the counter: worse.

FIELD RULE PEOPLE ACTUALLY USE: labeled NRR – 7. NIOSH on foam: about half the NRR, then – 7. Muffs hold up better (~75%, then – 7).

Fit beats the sticker. A seated NRR 29 beats a loose NRR 33. Roll, pull, hold — or the number on the box is fiction.